

Q1. Give below are two statements : One is labelled as Assertion (A) and the other is labelled Reason (R) :

{NEET 2023}

Assertion (A) :	The first stage of gametophyte in the life cycle of moss is protonema stage.
Reason (R):	Protonema develops directly from spores produced in capsule.

In the light of the above statements, choose the most appropriate answer from the options given below:

1.	(A) is not correct but (R) is correct.
2.	Both (A) and (R) are correct and (R) is the correct explanation of (A).
3.	Both (A) and (R) are correct but (R) is not the correct explanation of (A).
4.	(A) is correct but (R) is not correct.

Q2. Identify the pair of heterosporous pteridophytes among the following : {NEET 2023}

1. Equisetum and Salvinia
2. Lycopodium and Selaginella
3. Selaginella and Salvinia
4. Psilotum and Salvinia

Q3. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as

Reason (R):

{NEET 2023}

Assertion (A) :	In gymnosperms the pollen grains are released from the microsporangium and carried by air currents.
Reason (R):	Air currents carry the pollen grains to the mouth of the archegonia where the male gametes are discharged and pollen tube is not formed.

In the light of the above statements, choose the correct answer from the options given below:

1.	(A) is false but (R) is true.
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2	Both (A) and (R) are true and (R) is the correct explanation of (A).
3	Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
4	(A) is true but (R) is false.

Q4. Match List - I with List – II

{NEET 2023}

	List-I		List-II
(A)	Pteropsida	(I)	<i>Psilotum</i>
(B)	Lycopsida	(II)	<i>Equisetum</i>
(C)	Psilopsida	(III)	<i>Adiantum</i>
(D)	Sphenopsida	(IV)	<i>Selaginella</i>

Choose the correct answer from the options given below:

Options :	(A)	(B)	(C)	(D)
1.	II	III	I	IV
2.	III	I	IV	II
3.	II	III	IV	I
4.	III	IV	I	II

Q5. Which classes of algae possess pigment fucoxanthin and pigment phycoerythrin, respectively?

{NEET 2023}

1. Phaeophyceae and Chlorophyceae
2. Phaeophyceae and Rhodophyceae
3. Chlorophyceae and Rhodophyceae
4. Rhodophyceae and Phaeophyceae

Q6. Hydrocolloid carrageen is obtained from:

{NEET 2022}

1. Phaeophyceae only
2. Chlorophyceae and Phaeophyceae



3. Phaeophyceae and Rhodophyceae
4. Rhodophyceae only

Q7. Which of the following is incorrectly matched?

{NEET 2022}

1. *Volvox* - Starch
2. *Ectocarpus* - Fucoxanthin
3. *Ulothrix* - Mannitol
4. *Porphyra* - Floridian starch

Q8. Match the plant with the kind of life cycle it exhibits:

{NEET 2022}

List-I	List-II
(a) <i>Spirogyra</i>	(i) Dominant diploid sporophyte vascular plant, with highly reduced male or female gametophyte
(b) Fern	(ii) Dominant haploid free-living gametophyte
(c) <i>Funaria</i>	(ii) Dominant diploid sporophyte alternating with reduced gametophyte called prothallus
(d) <i>Cycas</i>	(iv) Dominant haploid leafy gametophyte alternating with partially dependent multicellular sporophyte

Choose the correct answer from the options given below:

	(a)	(b)	(c)	(d)
1.	(ii)	(iv)	(i)	(iii)
2.	(iv)	(i)	(ii)	(iii)
3.	(ii)	(iii)	(iv)	(i)
4.	(iii)	(iv)	(i)	(ii)

Q9. Match List -I with List-II

{NEET 2022}

List - I	List - II
a. <i>Cedrus</i>	(i) Pteridophyte
b. <i>Adiantum</i>	(ii) Gymnosperm
c. <i>Sphagnum</i>	(iii) Liverwort
d. <i>Marchantia</i>	(iv) Moss



	a.	b.	c.	d.
1.	(ii)	(iii)	(i)	(iv)
2.	(iii)	(i)	(iv)	(ii)
3.	(ii)	(i)	(iv)	(iii)
4.	(iii)	(iv)	(ii)	(i)

1. Primary Endospore Nucleus
2. Microspore Mother cell
3. Protonemal cell of a moss
4. Primary endosperm nucleus in dicot

{NEET 2022}

List-I	List-II
(a) <i>Chlamydomonas</i>	(i) Moss
(b) <i>Cycas</i>	(ii) Pteridophyte
(c) <i>Selaginella</i>	(iii) Alga
(d) <i>Sphagnum</i>	(iv) Gymnosperm

	(a)	(b)	(c)	(d)
1.	(iii)	(i)	(ii)	(iv)
2.	(iii)	(iv)	(ii)	(i)

3.	(iii)	(ii)	(i)	(iv)
4.	(ii)	(iii)	(i)	(iv)

Q12. Read the following statements and identify the characters related to the alga shown in the diagram:
{NEET 2022}

	a.	It is a member of Chlorophyceae.
	b.	Food is stored in the form of starch.
	c.	It is a monoecious plant showing oogonium and antheridium.
	d.	Food is stored in the form of laminarin or mannitol.
	e.	It shows dominance of pigments chlorophyll a, c and Fucoxanthin.

Choose the correct answer from the options given below :

1. (a) and (b) only
2. (a), (b) and (c) only
3. (a), (c) and (d) only
4. (c), (d) and (e) only

Q13. Which of the following algae contains mannitol as reserve food material?

{NEET 2021}

1.	<i>Volvox</i>	2.	<i>Ulothrix</i>
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Q18. Floridian starch has structure similar to :

{NEET 2020}

1. Amylopectin and glycogen
2. Mannitol and algin
3. Laminarin and cellulose
4. Starch and cellulose

Q19. Strobili or cones are found in:

{NEET 2020}

1.	<i>Pteris</i>	2.	<i>Marchantia</i>
3.	<i>Equisetum</i>	4.	<i>Salvinia</i>

Q20. Phycoerythrin is the major pigment in:

{NEET 2020}

1. Red algae
2. Blue-green algae
3. Green algae
4. Brown algae

Q21. Which of the following statements is incorrect about gymnosperms?

{NEET 2020}

1. They are heterosporous
2. Male and female gametophytes are free-living
3. Most of them have narrow leaves with thick cuticles
4. Their seeds are not covered

Q22. *Pinus* seeds cannot germinate and establish without fungal association. This is because:

{NEET 2019}



1. its seeds contain inhibitors that prevent germination.
2. its embryo is immature.
3. it has an obligate association with mycorrhizae.
4. it has a very hard seed coat.

Q23. From an evolutionary point of view, retention of the female gametophyte with developing young embryo on the parent sporophyte for some time, is first observed in:

{NEET 2019}

1.	Gymnosperms	2.	Liverworts
3.	Mosses	4.	Pteridophytes

(4) halophiles

Q24. Which one is *wrongly* matched?

{NEET 2018}

1. Uniflagellate gametes - *Polysiphonia*
2. Biflagellate zoospores - Brown algae
3. Gemma cups - *Marchantia*
4. Unicellular organism - *Chlorella*

Q25. Read the following five statements (I to V) and select the option with all correct statements.

{ NEET 2015}

I:	Mosses and lichens are the first organisms to colonise bare rock.
II:	<i>Selaginella</i> is a homosporous pteridophyte.
III:	Coralloid roots in <i>Cycas</i> have VAM.
IV:	Main plant body in bryophytes is gametophytic, whereas in pteridophytes it is sporophytic.



1. *Marchantia*
2. *Riccia*
3. *Funaria*
4. *Sphagnum*

Q30. An alga which can be employed as food for human beings:

{ AIPMT 2014}

1. *Ulothrix*
2. *Chlorella*
3. *Spirogyra*
4. *Polysiphonia*

Q31. Besides paddy fields, cyanobacteria are also found inside vegetative part of:

{ AIPMT 2013}

1.	<i>Cycas</i>	2.	<i>Equisetum</i>
3.	<i>Psilotum</i>	4.	<i>Pinus</i>

Q32. Isogamous condition with non-flagellated gametes are found in: { AIPMT 2013}

1. *Spirogyra*
2. *Volvox*
3. *Fucus*
4. *Chlamydomonas*

Q33. Monoecious plant of *Chara* shows the occurrence of:

{ AIPMT 2013}

1.	Stamen and carpel of the same plant
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2.	upper antheridium and lower oogonium on the same plant
3.	upper oogonium and lower antheridium on the same plant
4.	antheridiophore and archegoniophore on the same plant

Q36.What is common in all the three, Funaria, Dryopteris and Ginkgo? { NEET 2013}

1. Presence of archegonia
2. Well-developed vascular tissues
3. Independent gametophyte
4. Independent sporophyte

Q37. Which one of the following is wrongly matched?

{ NEET 2013}

1. *Spirogyra* -Motile gametes
2. *Sargassum* - Chlorophyll C
3. Basidiomycetes - Puffballs
4. *Nostoc*- Water blooms

Q38. *Cycas* and *Adiantum* resemble each other in having:

{AIMPT 2012}

1. Seeds
2. Motile sperms
3. Cambium
4. Vessels

Q39. Which one of the following is a correct statement?

{AIMPT 2012}

1.	Pteridophyte gametophyte has a protonemal and leafy stage
2.	In gymnosperms, female gametophyte is free-living
3.	Antheridiophores and protonema are present in pteridophytes
4.	Origin of seed habit can be traced in pteridophytes



Q40. Read the following five statements (I - V) and answer as asked next to them.
{AIMPT 2012}

I.	In <i>Equisetum</i> , the female gametophyte is retained on the parent sporophyte
II.	In <i>Ginkgo</i> , male gametophyte is not independent
III.	The sporophyte in <i>Riccia</i> is more developed than that in <i>Polytrichum</i>
IV.	Sexual reproduction in <i>Volvox</i> is isogamous
V.	The spores of slime molds lack cell walls

How many of the above statements are correct?

1. Three	2. Four
3. One	4. Two

Q41. Which one of the following pairs is wrongly matched?

{AIMPT 2012}

1. *Salvinia* - Prothallus
2. Viroids – RNA
3. Mustard-Synergids
4. *Ginkgo*-Archegonia

Q42. Compared with the gametophytes of the bryophytes, the gametophytes of vascular plants tends to be: **{AIMPT 2011}**

1. larger but to have smaller sex organs
2. larger and to have large sex organs
3. smaller and to have smaller sex organs
4. smaller but to have larger sex organs

Q43. The gametophyte is not an independent, free-living generation in:

{AIMPT 2011}



1. *Adiantum*
2. *Marchantia*
3. *Pinus*
4. *Polytrichum*

Q44. Archegoniophore is present in:

{AIMPT 2011}

1. *Chara*
2. *Adiantum*
3. *Funaria*
4. *Marchantia*

Q45. Mannitol is a sugar alcohol. It is as stored food in:

{AIPMT 2011}

1. *Fucus*
2. *Gracillaria*
3. *Chara*
4. *Porphyra*



ANSWER KEY

1. (2)	11. (2)	21. (2)	31. (1)	41. (1)
2. (3)	12. (2)	22. (3)	32. (1)	42. (4)
3. (4)	13. (3)	23. (4)	33. (3)	43. (3)
4. (4)	14. (1)	24. (1)	34. (2)	44. (4)
5. (2)	15. (2)	25. (3)	35. (3)	45. (1)
6. (7)	16. (2)	26. (4)	36. (1)	
7. (3)	17. (3)	27. (3)	37. (1)	
8. (3)	18. (1)	28. (4)	38. (2)	
9. (3)	19. (3)	29. (4)	39. (4)	
10. (3)	20. (1)	30. (2)	40. (3)	

